Test Series Code

09

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 22/03/2026

SYLLABUS

Physics:	Alternating Current, Electromagnetic Waves, Ray Optics and Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei + 30% Questions from cumulative syllabus of previously conducted all AITS test
Chemistry:	Bio-molecules, Purification and Analysis of Organic Compound, Coordination Compounds, The p-Block Elements, The d- and f- block Elements + 30% Questions from cumulative syllabus of previously conducted all AITS test
Botany:	Photosynthesis in Higher Plants, Plant Growth and Development, Organisms and Population, Ecosystem + 30% Questions from cumulative syllabus of previously conducted all AITS test
Zoology:	Biotechnology - Principles and Processes + 30% Questions from cumulative syllabus of previously conducted all AITS test

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. A Physics Wallah student steps out of his train compartment to get tea. Immediately after this, the train suddenly starts accelerating at $0.05\ \text{m/s}^2$. Unaware, the student spends 30 s waiting for the tea to be prepared, 18 s drinking it, and takes a 2 s reaction time upon realizing the train is moving. If he runs at a uniform speed, what is the minimum speed required to catch his same compartment door?

(1) 2.5 m/s
(2) 5.0 m/s
(3) 7.5 m/s
(4) 10.0 m/s

2. A parallel plate capacitor with a plate area of 100 cm^2 is being charged by a constant current of 4.0 mA. Consider a plane surface of area 25 cm^2 drawn parallel to the plates and situated symmetrically between them. What is the displacement current crossing this smaller area?

(1) 4.0 mA (2) 2.0 mA
(3) 1.0 mA (4) 0.5 mA

3. In a series LCR circuit, the voltage measured across the resistance, capacitor, and inductor are 30 V, 60 V, and 20 V respectively. What is the power factor of the circuit?

(1) 0.8 (2) 0.6
(3) 0.5 (4) 1.0

4. A purely inductive circuit is connected to an AC source. If the frequency of the source is doubled and the peak voltage is halved, the peak current in the circuit will become:

(1) Four times the original value
(2) Half the original value
(3) One-fourth the original value
(4) Same as the original value

5. An alternating voltage $V = 200\sin(100t)$ volts and t is in seconds, is applied to a series combination of a $100\,\mu\text{F}$ capacitor and a $100\,\Omega$ resistor. The phase difference between the current and the applied voltage is:

(1) 30° (2) 45°
(3) 60° (4) 90°

6. In a plane electromagnetic wave propagating in a vacuum, the peak electric field is 60 V/m. The average energy density in the magnetic field in J/m^3 is: (Take $\varepsilon_0 = 8.85\times10^{-12}\ \text{F/m}$)

(1) 7.96×10^{-9} (2) 1.59×10^{-8}
(3) 3.98×10^{-9} (4) 8.85×10^{-12}

7. Match the electromagnetic waves listed in **List-I** with their characteristic wavelength values provided in **List-II**.

	List-I (EM Wave)		**List-II (Wavelength)**
(A)	*X*-rays	(I)	500 nm
(B)	Microwaves	(II)	100 m
(C)	Visible light	(III)	1 nm
(D)	Radio waves	(IV)	1 cm

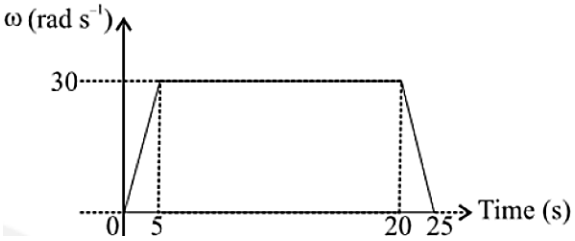
Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-IV, C-III, D-II

8. An equiconvex lens of focal length 20 cm in air is made of glass ($\mu = 1.5$). It is completely immersed in a liquid of refractive index 1.6. The new focal length and the nature of the lens will be:

(1) +160 cm, Converging
(2) –160 cm, Diverging
(3) –80 cm, Diverging
(4) +80 cm, Converging

9. The figure shows the angular velocity versus time graph of a flywheel. The angle in radian through which the flywheel turns during 25 s is:



(1) 120 (2) 480
(3) 600 (4) 750

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10. An astronomical telescope in normal adjustment has an objective lens of focal length 100 cm and an eyepiece of focal length 5cm. The magnifying power and the tube length of the telescope are respectively:

(1) 20 and 105 cm
(2) 20 and 95 cm
(3) 100 and 105 cm
(4) 0.05 and 100 cm

11. A point object is placed at a depth of 12 cm in water $\mu = \frac{4}{3}$. A transparent glass block $\mu = 1.5$ of thickness 3 cm is held flat on the water surface. The apparent depth of the object when viewed from normally above the glass block is:

(1) 11 cm (2) 9 cm
(3) 10 cm (4) 15 cm

12. A particle is dropped from a height *h*. If the coefficient of restitution of the particle with the ground is $\frac{1}{2}$, then the maximum height attained by the particle after three collisions with the ground is:

(1) $\frac{h}{8}$ (2) $\frac{h}{4}$
(3) $\frac{h}{64}$ (4) $\frac{h}{16}$

13. Two coherent monochromatic light beams of intensities *I* and 4*I* are superimposed. If the phase difference between the two beams at a certain point in the interference pattern is 60°, the resultant intensity at that point will be:

(1) 5*I* (2) 6*I*
(3) 7*I* (4) 9*I*

14. In a Young's double-slit experiment, the width of each individual slit is 0.20 mm and the distance between their centers is 1.60 mm. How many dark interference fringes will be observed within the central maximum of the diffraction pattern of an individual slit?

(1) 8 (2) 48
(3) 16 (4) 32

15. Unpolarized light of intensity I_0 passes through two polaroids P_1 and P_2. The transmission axis of P_2 makes an angle of 60° with the transmission axis of P_1. The intensity of the light transmitted through P_2 is:

(1) $\frac{I_0}{2}$ (2) $\frac{I_0}{4}$
(3) $\frac{I_0}{8}$ (4) $\frac{3I_0}{8}$

16. A beam of unpolarized light is incident from air onto a transparent medium of refractive index $\mu = \sqrt{3}$ at Brewster's angle. What is the angle between the direction of the incident ray and the refracted ray (angle of deviation)?

(1) 15° (2) 30°
(3) 45° (4) 60°

17. The work function of a metal is 4.0 eV. If monochromatic light of wavelength 200 nm falls on it, the maximum kinetic energy of the emitted photoelectrons is roughly: (Take $hc = 1240\,\text{eV}\cdot\text{nm}$)

(1) 2.2 eV (2) 6.2 eV
(3) 4.0 eV (4) 10.2 eV

18. An electron, a proton, and an alpha particle have the same kinetic energy. The correct order of their de Broglie wavelengths is:

(1) $\lambda_e > \lambda_p > \lambda_\alpha$ (2) $\lambda_\alpha > \lambda_p > \lambda_e$
(3) $\lambda_p > \lambda_e > \lambda_\alpha$ (4) $\lambda_e = \lambda_p = \lambda_\alpha$

19. Relate the experimental parameters of the Photoelectric Effect in **List-I** with the factors they primarily depend on in **List-II**.

List-I		List-II	
(A)	Saturation current	(I)	Depends on the Nature of material only
(B)	Stopping potential	(II)	Depends on Intensity of incident light
(C)	Work function	(III)	Depends on Frequency of light and Nature of material

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I
(2) A-III, B-II, C-I
(3) A-II, B-I, C-III
(4) A-I, B-III, C-II

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20. Two protons enter a region of transverse magnetic field. What will be the ratio of their angular speed, if their ratio of translational kinetic energy is $2\sqrt{2}:\sqrt{3}$?

(1) $\sqrt{3}:2\sqrt{2}$ (2) $3:2$

(3) $1:1$ (4) $2\sqrt{2}:\sqrt{3}$

21. A hydrogen atom in its ground state absorbs a photon of energy 12.09 eV. It then transitions back to the ground state. The maximum number of spectral lines that can be observed in the resulting emission spectrum is:

(1) 1 (2) 3

(3) 4 (4) 6

22. The ratio of the kinetic energy to the total energy of an electron in a stable Bohr orbit of the hydrogen atom is:

(1) $1:1$ (2) $1:-1$

(3) $1:2$ (4) $-1:2$

23. An alpha particle of kinetic energy $5\,\text{MeV}$ is projected head-on towards a stationary gold nucleus ($Z=79$). The distance of closest approach is roughly: (Take $\frac{1}{4\pi\varepsilon_0}=9\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2$ and $e=1.6\times10^{-19}\ \text{C}$)

(1) $4.5\times10^{-14}\,\text{m}$ (2) $2.5\times10^{-14}\,\text{m}$

(3) $8.0\times10^{-15}\,\text{m}$ (4) $1.2\times10^{-13}\,\text{m}$

24. Given below are two statements:

Statement I: According to the Bohr model, the angular momentum of an electron in a stable orbit of a hydrogen atom is quantized.

Statement II: Rutherford's model successfully explains why an electron revolving around the nucleus does not lose its energy and crash into the nucleus.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

25. The nuclear radius of a nucleus with mass number 125 is 5 fermi. The nuclear radius of a nucleus with mass number 64 will be:

(1) 4 fermi (2) 3 fermi

(3) 2.5 fermi (4) 3.2 fermi

26. In a certain nuclear fission reaction, the total mass of the reactants is $236.04\,\text{u}$ and the total mass of the products is $235.84\,\text{u}$. The energy released in the process is roughly: (Take $1\,\text{u}=931.5\,\text{MeV}$)

(1) 186.3 MeV (2) 200.0 MeV

(3) 93.15 MeV (4) 372.6 MeV

27. Two lighter nuclei combine to form a single heavier nucleus. In this process, the total mass of the initial nuclei is M_1 and the mass of the final nucleus is M_2. Which of the following relations is correct if the fusion process simultaneously releases energy?

(1) $M_1=M_2$ (2) $M_1>M_2$

(3) $M_1<M_2$ (4) $M_1\le M_2$

28. The binding energy per nucleon is maximum for the nucleus of:

(1) Uranium (^{238}U)

(2) Helium (^{4}He)

(3) Iron (^{56}Fe)

(4) Hydrogen (^{2}H)

29. Assuming the nuclear density of a nucleus of mass number A is 'ρ', the nuclear density of a nucleus of mass number $8A$ will be:

(1) 8ρ (2) 2ρ

(3) ρ (4) $\frac{\rho}{2}$

30. An alternating voltage given by $V(t)=200\sqrt{2}\sin(100t)$ Volts is applied across a heating coil of resistance 100 Ω. The coil is perfectly insulated from the surroundings but completely immersed in a large block of ice at 0°C. How much ice will melt in exactly 4.2 minutes? (Given L_f = 80 cal/g)

(1) 300 g (2) 420 g

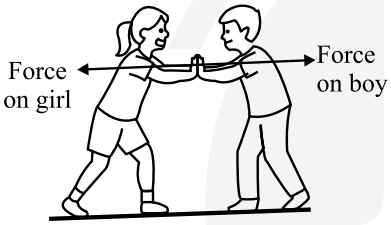
(3) 150 g (4) 600 g

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31. A horizontal plane mirror facing upwards is moving vertically up with a constant speed of 2 m/s. A small ball is dropped from rest from a height directly above the mirror. Assuming the ball has not yet hit the mirror, what is the magnitude of the relative velocity of the image of the ball with respect to the ball exactly 1 s after it is dropped? (Take $g = 10\ m/s^2$)

(1) 20 m/s
(2) 24 m/s
(3) 16 m/s
(4) 12 m/s

32. Two ice skaters, a man of mass 80 kg and a woman of mass 50 kg, are initially at rest on a flat ice bed. They push each other to gain some velocity. As a result, the man picks up a speed of 2.5 m/s. Consider the following statements:



Statement 1: The speed of the woman is $2.5\ m\ s^{-1}$.
Statement 2: The speed of the woman is $4.0\ m\ s^{-1}$.
Statement 3: Work done by the man while pushing the woman is 400 J.
Statement 4: Work done by the woman while pushing the man is less than the work done by the man pushing the woman.
Choose the **correct** options.

(1) Statements 1, 3 and 4 are only correct.
(2) Statements 1 and 4 only correct.
(3) Statements 2, 3 and 4 only correct.
(4) Statements 2 and 3 only correct.

33. A 220 V, 50 Hz AC source is connected to an ideal step-down transformer having a primary to secondary turns ratio of 10:1. If the secondary is connected to a 22 Ω resistor, what is the current in the primary coil?

(1) 0.1 A
(2) 1.0 A
(3) 10 A
(4) 0.01 A

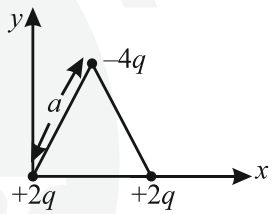
34. The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system?

(1) 20 Hz (2) 30 Hz
(3) 40 Hz (4) 10 Hz

35. The electric potential V in a certain one-dimensional region is given by the equation: $V(x) = ax^3 - bx + c$ (where a, b and c are positive constants, V is in volts, and x is in meters). How does the magnitude of the electric field, E, vary with position, x, in this region?

(1) It is constant
(2) It varies linearly
(3) It varies parabolically
(4) It varies cubically

36. Charge distribution at three vertices of an equilateral triangle of side a is shown in the figure. The net electric dipole moment of the system is:



(1) $2\sqrt{3}qa\hat{j}$
(2) $-2\sqrt{3}qa\hat{j}$
(3) $2\sqrt{3}qa\hat{i} + 2\sqrt{3}qa\hat{j}$
(4) $-2\sqrt{3}qa\hat{i} - 2\sqrt{3}qa\hat{j}$

37. Read the following statements and identify the option containing the **correct** statement (s).

(A) Metals, when heated to very high temperatures, exhibit perfect diamagnetism.
(B) The phenomenon of perfect diamagnetism in superconductors is called the Meissner effect.
(C) The magnetic susceptibility χ and relative magnetic permeability μ_r for a perfect diamagnetic material are 0 and –1 respectively.
(D) Superconducting magnets can be used in variety of situations like running magnetically levitated superfast trains.

(1) (A) and (B) only
(2) (B), (C) and (D) only
(3) (A), (B) and (D) only
(4) (B) and (D) only

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38. Two coils have a mutual inductance of 0.005 H. The current in the first coil changes according to the relation $i = 10\sin(100\pi t)$, where i is in amperes and t is in seconds. The maximum value of the EMF induced in the second coil is:

(1) 2π V (2) 5π V

(3) 10π V (4) 50π V

39. In a photoelectric setup, if the distance between a point light source and the photoelectric cell is halved, the saturation photoelectric current becomes:

(1) Halved

(2) Doubled

(3) Four times

(4) One-fourth

40. A particle is in uniform circular motion in the *x-y* plane. At a certain instant, its position vector is $\vec{r} = (3\hat{i} + 4\hat{j})$ m and its velocity vector is $\vec{v} = (-8\hat{i} + 6\hat{j})$ m/s. What is the acceleration vector $\vec{a}$ of the particle at this instant?

(1) $(-12\hat{i} - 16\hat{j})\,\text{m/s}^2$

(2) $(-16\hat{i} + 12\hat{j})\,\text{m/s}^2$

(3) $(12\hat{i} + 16\hat{j})\,\text{m/s}^2$

(4) $(16\hat{i} - 12\hat{j})\,\text{m/s}^2$

41. In a Young's Double Slit Experiment, a monochromatic light of wavelength 600 nm is used. The slit separation is 0.5 mm and the screen is 1 m away. A thin glass plate of thickness 1.5 μm and refractive index 1.5 is introduced in the path of one of the interfering beams. The central fringe shifts by:

(1) 1.5 mm (2) 3.0 mm

(3) 0.5 mm (4) 2.0 mm

42. The gravitational potential at the surface of a uniform solid sphere is –12 J/kg. What is the gravitational potential at the center of the sphere?

(1) –6 J/kg

(2) –18 J/kg

(3) –24 J/kg

(4) 12 J/kg

43. Given below are two statements:

Statement I: Optical fibers work on the principle of total internal reflection.

Statement II: For total internal reflection to take place, the angle of incidence in the denser medium must be strictly less than the critical angle.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

44. In an adiabatic expansion of one mole of an ideal gas, initial and final temperatures are T_1 and T_2 respectively, then the change in internal energy of the gas is:

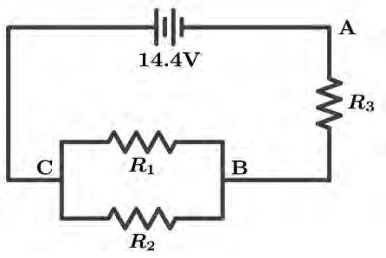
(1) $\frac{R}{\gamma - 1}(T_2 - T_1)$

(2) $\frac{R}{\gamma - 1}\left(\frac{T_1 T_2}{T_1 - T_2}\right)$

(3) $R(T_1 - T_2)$

(4) Zero

45. A battery of emf 14.4 V (negligible internal resistance) is connected to three resistors as shown with, $R_1 = R_2 = 20\,\Omega$ and $R_3 = 12\,\Omega$.



When a non-ideal voltmeter is connected across R_3, it reads 5.4 V. When the voltmeter is removed and a non-ideal ammeter is connected in series with R_2, it reads 240 mA. If both the voltmeter and ammeter are connected simultaneously in their respective positions, what will be their readings?

(1) 4.8 V and 320 mA

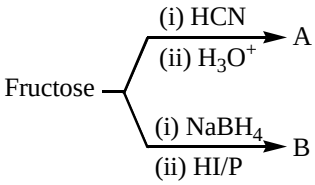
(2) 5.4 V and 240 mA

(3) 4.0 V and 300 mA

(4) 6.0 V and 200 mA

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46. The formula of A and B for the following reaction sequence are:



(1) A = $C_7H_{14}O_8$, B = C_6H_{14}
(2) A = $C_7H_{13}O_7$, B = $C_7H_{14}O$
(3) A = $C_7H_{12}O_8$, B = C_6H_{14}
(4) A = $C_7H_{14}O_8$, B = $C_6H_{14}O_6$

47. IUPAC name of $[Cr(H_2O)_5CO_3]Br$ is:
(1) pentaaquacarbonatochromate(III) bromide
(2) pentaaquacarbonatochromium(III) bromide
(3) pentaaquacarbonatochromium(II) bromide
(4) pentaaquacarbonatechromium(II) bromide

48. All the given elements show catenation property except:
(1) carbon (2) lead
(3) tin (4) silicon

49. The **correct** order of oxidising power of the given ions is:
(1) $MnO_4^- > Cr_2O_7^{2-} > VO_2^+$
(2) $VO_2^+ > Cr_2O_7^{2-} > MnO_4^-$
(3) $Cr_2O_7^{2-} > VO_2^+ > MnO_4^-$
(4) $Cr_2O_7^{2-} > MnO_4^- > VO_2^+$

50. The volume of 8 g of O_2 at STP is:
(1) 11.2 L (2) 5.6 L
(3) 22.4 L (4) 2.8 L

51. Given below are two statements:
Statement I: On hydrolysis proteins give rise to a large number of β-amino acids.
Statement II: Natural proteins are denatured by acids which convert into water soluble form of fibrous proteins to their water insoluble form.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

52. The volume of 0.1 M $AgNO_3$ solution required for complete precipitation of chloride ions present in 20 mL of 0.1 M solution of $[Cr(H_2O)_6]Cl_3$ is:
(1) 20 mL (2) 60 mL
(3) 10 mL (4) 40 mL

53. **Incorrect** statement regarding diamond is:
(1) each carbon is sp^3 hybridised
(2) it has high melting point
(3) thermodynamically most stable allotrope of carbon
(4) it is a covalent solid.

54. Given below are two statements:
Statement I: Scandium does not exhibit variable oxidation state.
Statement II: Cu^+ undergo disproportionation in aqueous solution.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

55. Isobars have same:
(1) mass number (2) number of neutrons
(3) atomic number (4) number of electrons

56. **Incorrect** statement among the following regarding enzymes is:
(1) Enzymes are biocatalysts
(2) Most enzymes are globular proteins
(3) Enzymes are very specific for a particular reaction
(4) Enzyme oxidase catalyses the hydrolysis of maltose into glucose

57. Given below are two statements:
Statement I: $[Fe(C_2O_4)_3]^{3-}$ is more stable complex ion than $[Fe(H_2O)_6]^{3+}$ complex ion.
Statement II: Wilkinson catalyst is used for the hydrogenation of alkenes.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

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58. Number of acidic oxides among the following is:
B_2O_3, Al_2O_3, Ga_2O_3, In_2O_3, Tl_2O_3
(1) 1 (2) 2
(3) 3 (4) 4

59. The transition metals form a large number of complex compounds due to:
(A) Comparatively smaller sizes of the metal ions
(B) Low ionic charge on metal
(C) Availability of d-orbitals on metal
Choose the **correct** option:
(1) A and B only
(2) A and C only
(3) B and C only
(4) A, B and C

60. Lowest oxidation state of halogens in their compound is:
(1) –3
(2) –1
(3) –7
(4) –2

61. Match **List-I** with **List-II**.

List-I (Vitamin)		List-II (Deficiency disease)	
(A)	Vitamin A	(I)	Beri-beri
(B)	Thiamine	(II)	Cheilosis
(C)	Ascorbic acid	(III)	Xerophthalmia
(D)	Riboflavin	(IV)	Scurvy

Choose the **correct** answer from the options given below:
(1) A-III; B-IV; C-II; D-I
(2) A-II; B-IV; C-I; D-III
(3) A-III; B-I; C-IV; D-II
(4) A-IV; B-I; C-II; D-III

62. The **correct** increasing order of field strength for the given ligands is:
(1) $SCN^- < NCS^- < S^{2-}$
(2) $SCN^- < S^{2-} < NCS^-$
(3) $NCS^- < SCN^- < S^{2-}$
(4) $NCS^- < S^{2-} < SCN^-$

63. Bond dissociation energy of E–H bond of the "H_2E" hydrides of group-16 elements follow order:
(1) O > S > Se > Te
(2) O > S > Te > Se
(3) S > O > Se > Te
(4) Te > Se > S > O

64. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Wacker process	(I)	MnO_2
(B)	Dry battery cells	(II)	$TiCl_4 + Al(CH_3)_3$
(C)	Ziegler catalysts	(III)	Cu/Ni alloy
(D)	'Silver' UK coins	(IV)	$PdCl_2$

Choose the **correct** answer from the options given below:
(1) A-IV; B-I; C-II; D-III
(2) A-II; B-IV; C-I; D-III
(3) A-III; B-I; C-IV; D-II
(4) A-III; B-IV; C-II; D-I

65. Mixture of petrol and diesel can be separated by:
(1) simple distillation.
(2) distillation under reduced pressure
(3) fractional distillation
(4) steam distillation

66. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Sucrose is a disaccharide and a non-reducing sugar.
Reason R: Sucrose involve glycosidic linkage between C_1 of β-glucose and C_2 of β-fructose.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

67. For the square planar complex [Mabcd], number of geometrical isomers is:
(1) 0 (2) 1
(3) 2 (4) 3

68. Which one has highest boiling point?
(1) He
(2) Ne
(3) Kr
(4) Xe

69. Which among the following metals has highest enthalpy of atomisation?
(1) V (2) Mn
(3) Ni (4) Fe

[Contd...

70. The vapour pressure of 1.5 molal dilute aqueous solution of sucrose at 38°C is nearly:
(Given: vapour pressure of H_2O at 38°C is 50 mm of Hg)
(1) 49.25 mm of Hg
(2) 49.85 mm of Hg
(3) 48.15 mm of Hg
(4) 48.65 mm of Hg

71. Sulphur containing amino acids among the following are:
(A) Isoleucine
(B) Cysteine
(C) Lysine
(D) Methionine
(1) A and C
(2) A and B
(3) B and D
(4) C and D

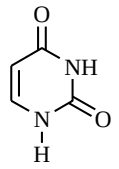
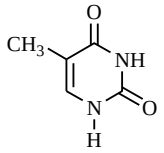
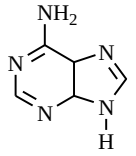
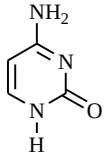
72. The **correct** statements regarding complex $[Co(NH_3)_6]^{3+}$ are:
(A) Diamagnetic in nature
(B) Octahedral in shape
(C) High spin Complex
(1) A and B only
(2) B and C only
(3) A and C only
(4) A, B and C

73. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Among hydrogen halides, HF has highest boiling point.
Reason R: There is strong H-bonding between HF molecules.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

74. **Incorrect** statement regarding lanthanoids is:
(1) Ce^{4+} is a good analytical reagent
(2) Eu^{2+} is a strong reducing agent.
(3) Tb^{4+} is an oxidant
(4) Yb^{2+} is a oxidant

75. For which of the following cell, $E^{o}_{cell} = 0$?
(1) $Ag(s)\,|\,Ag^{+}(0.1\,M)\,||\,Ag^{+}(0.2\,M)\,|\,Ag(s)$
(2) $Fe^{2+}(0.1\,M)\,|\,Fe^{3+}(0.2\,M)\,||\,Ag^{+}(0.1\,M)\,|\,Ag(s)$
(3) $Fe^{2+}(0.2\,M)\,|\,Fe^{3+}(0.1\,M)\,||\,I_2(s)\,|\,I^{-}(0.1\,M)$
(4) $Ni(s)\,|\,Ni^{2+}(0.1\,M)\,||\,Ag^{+}(0.1\,M)\,|\,Ag(s)$

76. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Adenine	(I)	
(B)	Cytosine	(II)	
(C)	Thymine	(III)	
(D)	Uracil	(IV)	

Choose the **correct** answer from the options given below:
(1) A-IV; B-I; C-II; D-III
(2) A-II; B-IV; C-I; D-III
(3) A-III; B-IV; C-II; D-I
(4) A-III; B-I; C-IV; D-II

77. Optically active species among the following is:
(1) cis-$[CrCl_2(ox)_2]^{3-}$
(2) trans-$[PtCl_2(en)_2]^{2+}$
(3) cis-$[Co(NH_3)_4Cl_2]^{+}$
(4) trans-$[Co(NH_3)_4Cl_2]^{+}$

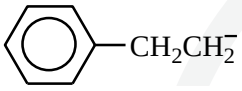
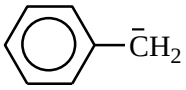
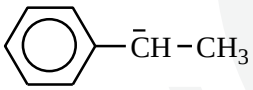
78. Positive lassaigne's test for nitrogen is given by:
(1) NH_2OH
(2) CH_3CN
(3) NH_3
(4) N_2H_4

79. Iodination of methane is:
(1) very fast and reversible reaction
(2) very slow and reversible reaction
(3) very slow irreversible reaction
(4) very fast and irreversible reaction

[Contd...

80. The adsorbent used in the adsorption chromatography is/are:
(A) Silica gel
(B) Alumina
(C) Quick lime
(D) Magnesia
(1) C and D
(2) A and D
(3) B and D
(4) A and B

81. Which of the following vitamins belongs to water soluble vitamins?
(1) Vitamin E
(2) Vitamin K
(3) Vitamin C
(4) Vitamin D

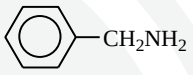
82. Most stable carbanion among the following is:
(1) 
(2) 
(3) 
(4) 

83. 0.5 g of an organic compound is digested as per Kjeldahl's method and evolved ammonia is absorbed in 100 ml of $\frac{M}{10}$ H_2SO_4. The unreacted acid requires 200 ml of $\frac{M}{20}$ NaOH for complete neutralisation. The percentage of nitrogen in organic compound is:
(1) 7%
(2) 14%
(3) 28%
(4) 34%

84. Number of monochloro derivative (excluding stereo isomers) for 1,4-dimethyl cyclohexane is:
(1) 2 (2) 3
(3) 4 (4) 5

85. Given below are two statements:
Statement I: Picric acid is mono carboxylic acid.
Statement II: Phenol-2,4-disulphonic acid is treated with conc. HNO_3 to get picric acid.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

86. A mixture of benzaldehyde and formaldehyde on heating with aq. NaOH solution gives:
(1) benzyl alcohol and sodium formate
(2) sodium benzoate and methyl alcohol
(3) sodium benzoate and sodium formate
(4) benzyl alcohol and methyl alcohol

87. Most basic compound in vapour phase among the following is:
(1) CH_3NH_2
(2) 
(3) $(CH_3)_2NH$
(4) $(CH_3)_3N$

88. The specific rate constant of a first order reaction depends on:
(A) pressure
(B) temperature
(C) concentration of reactant
(D) presence of catalyst
(1) A and B
(2) B and C
(3) C and D
(4) B and D

89. 2 mol of A (g) are introduced into a 2 L flask which dissociates as:
$A(g) \rightleftharpoons B(g) + C(g)$,
at equilibrium, 0.5 mol of B (g) are present. The equilibrium constant of the reaction is:
(1) 0.012
(2) 0.111
(3) 0.083
(4) 0.234

90. Methods used for purification of organic compounds are based on:
(1) nature of compound only
(2) presence of impurity only
(3) nature of compound and presence of impurity
(4) neither on nature of compound nor on presence of impurity.

[Contd...

91. During decomposition, _____ leads to accumulation of a dark coloured amorphous substance which is highly resistant to microbial action.
(1) leaching
(2) fragmentation
(3) mineralisation
(4) humification

92. When the number of reproductive individuals is more than that of pre- and post-reproductive individuals, the population is said to be;
(1) increasing.
(2) declining.
(3) stable.
(4) extinct.

93. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Removal of shoot tips in tea plantations results in the growth of lateral buds.
Reason (R): The growing apical bud inhibits the growth of the lateral (axillary) buds, a phenomenon called heterophylly.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

94. In non-cyclic photophosphorylation, plastoquinone is;
(1) the primary acceptor of electrons from PS II.
(2) the primary acceptor of electrons from PS I.
(3) the reduced form of plastoquinol.
(4) a proton carrier.

95. In the three-domain system, the eukaryotes have been classified into how many domains?
(1) One
(2) Two
(3) Three
(4) Zero

96. Match **List-I** with **List-II** w.r.t a typical grassland ecosystem.

List-I		List-II	
(A)	First Trophic level	(I)	Tree
(B)	Second Trophic level	(II)	Grasshopper
(C)	Third Trophic level	(III)	Bird
(D)	Fourth Trophic level	(IV)	Lion

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-III, C-IV, D-I
(2) A-II, B-IV, C-III, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-III, B-IV, C-I, D-II

97. The ability of a plant to follow different pathways in response to environment or phases of life to form different kinds of structures is called:
(1) Differentiation.
(2) Plasticity.
(3) Development.
(4) Maturation.

98. In an area having a single banyan tree and 200 *Parthenium* plants, the measure of the population size is more meaningful in terms of;
(1) per cent cover or biomass.
(2) number of individuals.
(3) number of fruits borne by them.
(4) number of parasites on them.

99. Given below are two statements:
Statement I: Photorespiration does not occur in C_4 plants.
Statement II: C_4 plants have a mechanism that decreases the concentration of CO_2 at the enzyme site of RuBisCO.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

100. The rate of formation of new organic matter by consumers is called:
(1) Gross primary productivity.
(2) Net primary productivity.
(3) Primary production.
(4) Secondary productivity.

101. The plant growth regulator that acts as a general plant growth inhibitor and an inhibitor of plant metabolism is;
(1) Auxin.
(2) Gibberellin.
(3) Cytokinin.
(4) Abscisic acid

102. Read the given statements regarding the *lac* operon.
A. The *i* gene codes for the repressor of the *lac* operon.
B. The *z* gene codes for beta-galactosidase.
C. The *y* gene codes for transacetylase.
D. The *a* gene codes for permease.
E. The products of all the three structural genes are required for the metabolism of lactose.
The **correct** statements are;
(1) A, C, E
(2) B, D
(3) A, B, E
(4) A, B, C, D, E

103. An example of commensalism is;
(1) an orchid growing as an epiphyte on a mango branch.
(2) the association between fungi and the roots of higher plants.
(3) the interaction between a tiger and a deer.
(4) cuckoo laying its eggs in a crow's nest.

104. Identify the **false** statement about the organism shown below.



(1) In an ecosystem, they act as decomposers.
(2) They help in fragmentation of detritus.
(3) These are heterotrophic organisms.
(4) They perform enzymatic catabolism of organic substances.

105. Given below are two statements:
Statement I: The liverworts reproduce sexually by fragmentation.
Statement II: The gametes in liverworts are called gemmae.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

106. During photosynthesis, water splitting and ATP synthesis occur in;
(1) stroma of chloroplast.
(2) thylakoid lumen.
(3) thylakoid lumen and stroma of chloroplast, respectively.
(4) stroma of chloroplast and thylakoid lumen, respectively.

107. In a monocot root;
(1) there are usually more than six xylem bundles.
(2) pith is small or inconspicuous.
(3) the vascular bundles are conjoint.
(4) secondary growth is observed.

108. Given below are two statements:
Statement I: The pyramid of biomass in the sea is generally inverted.
Statement II: The pyramid of energy is always inverted.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

109. The *Verhulst-Pearl Logistic Growth* is described by the equation:
(1) $\frac{dN}{dt} = rN\left(\frac{N-K}{K}\right)$
(2) $\frac{dN}{dt} = rN\left(\frac{K-N}{N}\right)$
(3) $\frac{dN}{dt} = rN\left(\frac{N-K}{N}\right)$
(4) $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

110. How many of the given physiological effects are correct for ethylene?

A. Promotes senescence and abscission of plant organs.
B. Induces seed and bud dormancy.
C. Promotes rapid internode/petiole elongation in deep water rice plants.
D. Promotes root growth and root hair formation.
E. Induces flowering in mango.

Choose the **correct** option.

(1) Two only
(2) Three only
(3) Four only
(4) Five

111. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): PS I and PS II are named in sequence of their function during light reaction.

Reason (R): PS I was discovered before PS II.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

112. Identify the **correctly** matched pair.

(1) Parietal placentation - *Argemone*
(2) Axile placentation - Pea
(3) Marginal placentation - *Dianthus*
(4) Free central placentation - China rose

113. Match **List-I** with **List-II**.

List-I (Organism)		**List-II (Kingdom)**	
(A)	Mycoplasma	(I)	Fungi
(B)	*Plasmodium*	(II)	Protista
(C)	*Ginkgo*	(III)	Plantae
(D)	*Neurospora*	(IV)	Monera

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

114. Father of Ecology in India is;

(1) Ramdeo Misra.
(2) G. N. Ramachandran.
(3) Panchanan Maheshwari.
(4) Har Gobind Khorana.

115. To improve the shape of apples, the PGR that must be sprayed on them is;

(1) Auxin.
(2) Gibberellin.
(3) Ethylene.
(4) Abscisic acid.

116. Identify the **correct** statements regarding competition.

A. Competition occurs when closely related species compete for different resources.
B. Totally unrelated species could also compete for the same resource.
C. Resources should never be limiting for competition to occur.
D. In interference competition, the feeding efficiency of one species is reduced by presence of the other.
E. Herbivores and plants are less adversely affected by competition than carnivores.

Choose the **correct** option.

(1) A and C only
(2) B and D only
(3) B, C and E only
(4) A, D and E only

117. Match **List-I** with **List-II**.

List-I (Process/Pathway)		**List-II (One of the Products)**	
(A)	Light reaction	(I)	Ethanol
(B)	Glycolysis	(II)	Oxygen
(C)	Fermentation in yeast	(III)	Phosphoglycolate
(D)	Photorespiration	(IV)	Pyruvate

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-IV, C-I, D-III

118. The degradation product of fatty acid that is also its precursor for synthesis;

(1) is a 3-carbon compound.
(2) is also a product of lactic acid fermentation.
(3) enters the Krebs' cycle during aerobic respiration.
(4) is the primary acceptor of CO_2 in C_4 plants.

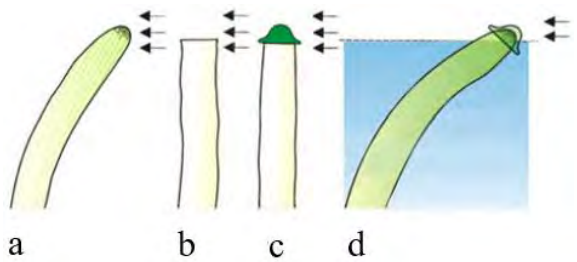
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119. Arrange the following events of meiotic cell cycle in the correct sequence:
A. Formation of dyad of cells
B. Pairing of homologous chromosomes
C. Formation of 4 haploid cells
D. Duplication of DNA
E. Splitting of centromere
Choose the **correct** option.
(1) D → B → A → E → C
(2) A → B → C → D → E
(3) E → C → B → A → D
(4) B → E → D → C → A

120. Consider the following statements.
A. Sun is unexceptionally the only source of energy for all ecosystems on Earth.
B. Plants capture 2-10 per cent of the photosynthetically active radiation (PAR).
C. The flow of energy from the sun to producers and then to consumers is bidirectional.
D. Ecosystems are not exempt from the Second Law of thermodynamics.
E. The transfer of energy follows the 10 per cent law in the grazing food chain.
Choose the **correct** option.
(1) A, C, E are correct
(2) A, B, C are correct
(3) C, D, E are correct
(4) B, D, E are correct

121. In red algae;
(1) gametes are motile.
(2) asexual spores are motile.
(3) complex post-fertilisation developments take place.
(4) sexual reproduction is isogamous type.

122. In the following experiment performed by Charles Darwin and Francis Darwin, they observed the phenomenon of;



(1) vernalisation.
(2) phototropism.
(3) dedifferentiation.
(4) heterophylly.

123. Evaluate the following based on post-fertilisation fate of given structures in angiosperms.
A. Ovule → Fruit
B. Ovary → Seed
C. Integument → Seed coat
D. Central cell → Zygote
E. Antipodals → Degenerate
The **correct** ones are;
(1) C and E
(2) A, B and E
(3) C and D
(4) A, C and D

124. Which of the following taxonomic categories is same for monkey, cat, human and housefly?
(1) Order
(2) Species
(3) Class
(4) Kingdom

125. Given below are two statements:
Statement I: The two chains of a DNA double helix have anti-parallel polarity.
Statement II: In a DNA double helix, a purine always comes opposite to a pyrimidine.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

126. Select the **correctly** matched pair w.r.t a pond ecosystem.
(1) Primary consumer - Bacteria
(2) Producer - Algae
(3) Secondary consumer - Flagellate
(4) Decomposer - Submerged plant

127. Match **List-I** with **List-II**.

List-I (Scientist)		**List-II (Associated with PGR)**	
(A)	F. W. Went	(I)	Gibberellin
(B)	E. Kurosawa	(II)	Ethylene
(C)	Miller et al.	(III)	Auxin
(D)	H. H. Cousins	(IV)	Cytokinin

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-IV, C-I, D-III

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128. Identify the **correct** statement about the dark reaction of photosynthesis.

(1) They can only occur in dark.

(2) They occur in the stroma of chloroplast.

(3) They require NADPH but not ATP.

(4) Temperature has no effect on them.

129. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Predators in nature are 'prudent'.

Reason (R): Predators act as 'conduits' for energy transfer across trophic levels.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

130. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Standing crop	(I)	Rate of biomass production
(B)	Productivity	(II)	Vertical distribution of species occupying different levels in an ecosystem
(C)	Stratification	(III)	Mass of living material at a particular time at a trophic level
(D)	Decomposition	(IV)	Breakdown of complex organic matter into inorganic substances

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-I, C-II, D-IV

(2) A-III, B-II, C-I, D-IV

(3) A-IV, B-I, C-II, D-III

(4) A-II, B-IV, C-III, D-I

131. The number of microtubules in a centriole is;

(1) 27.

(2) 20.

(3) 29.

(4) 18.

132. The ovary is said to be superior in;

(1) plum.

(2) peach.

(3) rose.

(4) china rose.

133. _____ proposed that plants change light energy to chemical energy by transferring an electron in an organised array of pigment molecules and other substances.

(1) Joseph Priestley

(2) Jan Ingenhousz

(3) Melvin Calvin

(4) Julius von Sachs

134. In a chromatogram, the colour yellow to yellow-orange is shown by;

(1) Chlorophyll *a*.

(2) Chlorophyll *b*.

(3) Xanthophylls.

(4) Carotenoids.

135. Identify the correct statements w.r.t formation of a tracheary element from a meristematic cell.

A. The cells undergo few to major structural changes.

B. The process is called dedifferentiation.

C. The cells would lose their protoplasm.

D. The cells develop an inelastic secondary cell wall.

E. The cells develop a very strong, lignocellulosic secondary cell walls.

Choose the **correct** option.

(1) A, C, E

(2) C, D, E

(3) A, B, E

(4) B, C, D

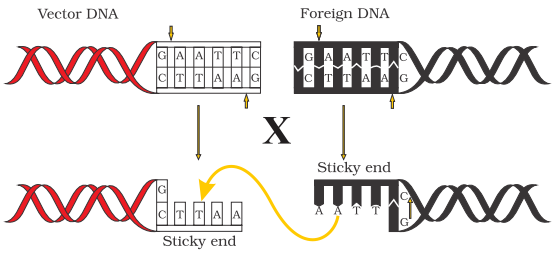
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136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: DNA cannot pass through cell membranes even if the membrane is made permeable by chemical treatment.
Reason R: DNA is a hydrophilic molecule, while the core of the plasma membrane is hydrophobic.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

137. Regarding the pBR322 vector, consider the following statements.
A. The 'rop' gene codes for the proteins involved in the replication of the plasmid.
B. If a foreign DNA is ligated at the *Bam*H I site, the recombinant plasmid loses ampicillin resistance.
C. The ampicillin resistance gene has recognition sites for *Pst* I and *Pvu* I.
D. Kanamycin resistance is used as a selectable marker to differentiate transformants from non-transformants.
Which of the above statements are **correct**?
(1) A, B and C (2) A and C
(3) B, C and D (4) A and D

138. In which of the following animal groups is a water vascular system found, and what is its role?
(1) Porifera; intracellular digestion
(2) Echinodermata; capture and transport of food
(3) Mollusca; excretion of waste
(4) Ctenophora; bioluminescence

139.



What will be the **most** likely result if X is not kept same for vector and foreign DNA?
(1) A highly efficient recombinant DNA molecule will form.
(2) DNA ligase will join them even without complementary sticky ends.
(3) The DNA fragments will not join because the sticky ends are not complementary.
(4) The experiment will produce more "blue" colonies.

140. Match **List-I** with **List-II**.

List-I		List-II	
(A)	P-wave	(I)	Repolarisation of the ventricles
(B)	QRS complex	(II)	Volume of blood pumped out by each ventricle per minute
(C)	T-wave	(III)	Depolarisation of the atria
(D)	Cardiac output	(IV)	Depolarisation of the ventricles

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-II, C-III, D-IV

141. Which of the following describes the immediate consequence of activation of JG cells?
(1) Release of Atrial Natriuretic Factor (ANF) which causes vasodilation.
(2) Release of renin which converts angiotensinogen in blood to angiotensin I.
(3) Suppression of ADH release from the neurohypophysis.
(4) Direct stimulation of the adrenal cortex to release aldosterone.

142. Read the following statements regarding the mechanism of muscle contraction:
A. Muscle contraction is initiated by a signal sent by the CNS via a sensory neuron.
B. A neural signal reaching the neuromuscular junction releases the neurotransmitter acetylcholine, generating an action potential in the sarcolemma.
C. Increase in Ca^{2+} level leads to the binding of calcium with a subunit of troponin on actin filaments.
D. Utilising the energy from ATP hydrolysis, the myosin head binds to the exposed active sites on actin to form a cross bridge.
E. The 'Z' lines attached to the actins are pulled outwards, thereby causing a lengthening of the sarcomere.
How many of the above statements are **correct**?
(1) Two (2) Three
(3) Four (4) Five

143. The layer of the cranial meninges that is in direct contact with the brain tissue is known as:
(1) Dura mater (2) Arachnoid
(3) Pia mater (4) Corpus callosum

[Contd...

144. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Thyrocalcitonin	(I)	Hypoglycemic hormone
(B)	Parathyroid hormone	(II)	Main mineralocorticoid in the human body
(C)	Insulin	(III)	Increases blood Ca^{2+} levels
(D)	Aldosterone	(IV)	Decreases blood Ca^{2+} levels

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-III, C-I, D-II
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-I, C-III, D-II

145. Given below are two statements:

Statement I: Oogenesis is initiated during the embryonic development stage when a couple of million oogonia are formed within each fetal ovary.

Statement II: Each ovary is covered by a thick epithelium which encloses the ovarian stroma.

In the light of above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

146. Under the Medical Termination of Pregnancy (Amendment) Act, 2017, what is the maximum gestational age up to which a pregnancy can be terminated?

(1) 12 weeks
(2) 20 weeks
(3) 24 weeks
(4) 28 weeks

147. Which of the following scenarios **best** explains the concept of convergent evolution?

(1) The forelimbs of whales, bats, Cheetah, and human
(2) Darwin's finches radiating from a seed-eating ancestor to insectivorous and vegetarian forms
(3) The eye of the octopus and the eye of mammals
(4) The appearance of dark-winged moths in England post-industrialisation

148. Identify the **mismatched** pair:

(1) Micro-injection: Direct injection of recombinant DNA into the nucleus of an animal cell.
(2) Biolistics: Bombardment of plant cells with high-velocity micro-particles of gold or tungsten.
(3) Disarmed pathogen: Using a virus or bacterium to infect a cell without causing disease.
(4) Heat shock: Treating cells with ice followed by 100°C to force DNA uptake.

149. What happens when a recombinant DNA is inserted within the coding sequence of beta-galactosidase?

(1) The colonies produce a bright blue colour.
(2) The colonies do not produce any colour and are identified as recombinant.
(3) The colonies produce a red colour indicating successful transformation.
(4) The bacteria die due to lack of antibiotic resistance.

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: For the multiplication of any alien piece of DNA in an organism, it needs to be a part of a chromosome which has a specific sequence known as 'origin of replication'.

Reason R: Origin of replication is a sequence from where replication starts and it also controls the copy number of the linked DNA.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

151. If you ligate a foreign DNA at the *Sal* I site of pBR322, the transformants will be able to grow on:

(1) Medium containing both ampicillin and tetracycline.
(2) Medium containing ampicillin but not on medium containing tetracycline.
(3) Medium containing tetracycline but not on medium containing ampicillin.
(4) Neither ampicillin nor tetracycline medium.

152. Which of the following groups contains only nucleosides?

(1) Adenine, Guanine, Cytosine, Uracil
(2) Adenosine, Guanosine, Thymidine, Uridine
(3) Adenylic acid, Thymidylic acid, Guanylic acid
(4) Adenosine, Cytosine, Uridylic acid, Thymine

[Contd...

153. Consider the following statements w.r.t cockroach.
A. The head is formed by the fusion of six segments and shows great mobility in all directions due to a flexible neck.
B. Tegmina are opaque, dark, and leathery, covering the hind wings when at rest.
C. The abdomen in females and males consists of 10 and 9 segments respectively.
D. In males, the genital pouch lies at the hind end of the abdomen bounded dorsally by the 9th and 10th terga and ventrally by the 9th sternum.
E. Males bear a pair of short, thread-like anal styles which are absent in females.
Which of the above statements are **correct**?
(1) A, B, C and D (2) A, B, D and E
(3) B, C, D and E (4) A, C, D and E

154. Downstream processing includes:
(1) Separation and Purification
(2) Gene expression and Ligation
(3) Denaturation and Annealing
(4) Transformation and Screening

155. In the PCR technique, the step of Annealing requires which of the following?
(1) High temperature
(2) Thermostable DNA polymerase and deoxynucleotides.
(3) Two sets of primers
(4) Chilled ethanol to precipitate DNA.

156. Which of the following is **not** a feature required to facilitate cloning into a vector?
(1) Origin of replication (ori)
(2) Selectable marker
(3) Presence of multiple recognition sites for the same restriction enzyme
(4) Cloning sites

157. Which enzyme is used to join the 'sticky ends' of the DNA fragments?
(1) Restriction endonuclease
(2) DNA polymerase
(3) DNA ligase
(4) Exonuclease

158. Given below are two statements:
Statement I: The tumor-inducing (Ti) plasmid of *Agrobacterium tumifaciens* has been modified into a cloning vector which is no longer pathogenic.
Statement II: Retroviruses in animals have the ability to transform normal cells into cancerous cells and are used as vectors after being disarmed.
In the light of above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

159. Which of the following molecules moves farthest in an agarose gel during electrophoresis?
(1) A 100 bp DNA fragment
(2) A 1000 bp DNA fragment
(3) A circular plasmid of 5000 bp
(4) A 200 bp DNA fragment

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In human females, the sex of the baby is determined by the father and not by the mother.
Reason R: All haploid gametes produced by the male have the sex chromosome X, whereas the female gametes can carry either the X or Y chromosome.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

161. In the simple stirred-tank bioreactor, the foam breaker is located:
(1) At the very bottom near the sterile air inlet.
(2) Around the motor to prevent overheating.
(3) At the top of the culture broth surface.
(4) Inside the steam inlet for sterilization.

162. Given below are two statements:
Statement I: About 2000 million years ago, the first cellular forms of life appeared on earth.
Statement II: The Neanderthal man with a brain size of 1400cc lived in near east and central Asia between 1,00,000-40,000 years back.
In the light of above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

163. A patient presents with sustained high fever (39 to 40 °C), weakness, stomach pain, constipation, and loss of appetite. Which diagnostic test would a doctor **most** likely prescribe to confirm the suspected bacterial disease?
(1) ELISA test (2) Widal test
(3) MRI (4) Biopsy

164. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Gel electrophoresis	(I)	Construction of first rDNA molecule
(B)	Polymerase Chain Reaction (PCR)	(II)	Synthesizing multiple copies of the gene of interest *in vitro.*
(C)	*Salmonella typhimurium*	(III)	Autonomously replicating circular extra-chromosomal DNA)
(D)	Plasmid	(IV)	Separating DNA fragments based on their size under an electric field.

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-IV, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-II, B-IV, C-III, D-I
(4) A-IV, B-II, C-I, D-III

165. Pathogenic bacteria gain resistance to antibiotics due to changes in their:
(1) Nucleoid (2) Cytoplasm
(3) Plasmids (4) Nucleus

166. In the nomenclature of the enzyme *EcoR* I, the Roman numeral 'I' indicates:
(1) The order of its discovery in the strain of bacteria.
(2) The number of recognition sites it has in pBR322.
(3) The size of the enzyme.
(4) The first letter of the genus.

167. Which of the following describes the **correct** molecular nature and function of the Ribozyme?
(1) It is a proteinaceous enzyme that catalyzes the breaking of peptide bonds.
(2) It is a nucleic acid (RNA) that behaves like an enzyme and has catalytic power.
(3) It is an inorganic catalyst functioning optimally at high temperature and pressure.
(4) It is a lipid derivative that lowers the activation energy in fat metabolism.

168. Identify the **correctly** matched pair:
(1) Exonucleases: Make cuts at specific positions within the DNA.
(2) Endonucleases: Remove nucleotides from the ends of the DNA.
(3) Palindrome: Sequence of base pairs that reads the same on the two strands when orientation of reading is kept the same.
(4) Okazaki fragments: Specific recognition sequence identified by endonucleases

169. When a protein-encoding gene is expressed in a heterologous host, the resulting protein is referred to as a:
(1) Recombinant protein
(2) Primary metabolite
(3) Secondary metabolite
(4) Native protein

170. The type of epithelium found in the walls of blood vessels and air sacs of lungs is:
(1) Cuboidal (2) Columnar
(3) Squamous (4) Ciliated

171. Match **List-I** with **List-II**.

List-I		List-II	
(A)	T-DNA	(I)	Natural polymer extracted from sea weeds
(B)	Bioreactor	(II)	Transform normal plant cells into a tumor
(C)	Vector	(III)	Large scale production
(D)	Agarose	(IV)	Plasmid or Virus

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-III, B-II, C-I, D-IV
(3) A-II, B-IV, C-III, D-I
(4) A-IV, B-I, C-II, D-III

172. Given below are two statements:
Statement I: *Taq* polymerase is used in the denaturation step of PCR.
Statement II: It is a thermostable DNA polymerase isolated from the bacterium *Thermus aquaticus*.
In the light of above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I is correct but Statement II is incorrect.

173. Continuous culture systems are used because:
(1) They produce a larger biomass and higher yields of desired protein.
(2) They are easier to clean than batch cultures.
(3) They require no oxygen.
(4) They only use plant cells.

174. To visualize DNA on an agarose gel, it must be stained with _____ and exposed to _____.
(1) Methylene blue, Visible light
(2) Ethidium bromide, UV radiation
(3) Acetocarmine, Infrared rays
(4) Iodine, UV radiation

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	EFB	(I)	Making multiple identical copies of any template DNA
(B)	Cloning	(II)	Maintenance of sterile (microbial contamination-free) ambience
(C)	Bioprocess engineering	(III)	European Federation of Biotechnology
(D)	Sexual reproduction	(IV)	Provides opportunities for variations

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-III, B-I, C-II, D-IV
(3) A-III, B-II, C-I, D-IV
(4) A-I, B-II, C-III, D-IV

176. Which of the following is **not** a cloning vector?
(1) Probe
(2) Ti plasmid of *Agrobacterium*
(3) Reterovirus
(4) pBR322

177. Spooling is:
(1) Cutting of separated DNA bands from the agarose gel
(2) Transfer of separated DNA fragments to synthetic membranes
(3) Collection of isolated DNA
(4) Amplification of DNA

178. Given below are two statements:
Statement I: DNA is insoluble in cold ethanol and precipitates out as fine threads.
Statement II: The first recombinant DNA was constructed by Stanley Cohen and Herbert Boyer in 1972.
In the light of above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Recombinant DNA technology is also called Genetic Engineering.
Reason R: It involves techniques to alter the chemistry of genetic material and introduce these into host organisms.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

180. *Hind* II the first restriction endonuclease discovered, always cuts DNA molecules at a specific recognition sequence consisting of:
(1) 4 bp
(2) 10 bp
(3) 8 bp
(4) 6 bp

■■■

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